

Applied Statistical Analysis I / Quantitative Methods I

Problem Set 1

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Problem 1

Question 1.1: 90% Confidence Interval

First, enter the sample and calculate the sample size, mean, and standard deviation.

```
y <- c(105, 69, 86, 100, 82, 111, 104, 110, 87, 108,  
       87, 90, 94, 113, 112, 98, 80, 97, 95, 111,  
       114, 89, 95, 126, 98)
```

```
length(y)
```

```
## [1] 25
```

```
mean(y)
```

```
## [1] 98.44
```

```
sd(y)
```

```
## [1] 13.09287
```

The sample size is

$$n = 25$$

with sample mean

$$\bar{x} = 98.44$$

and sample standard deviation

$$s = 13.09.$$

Because the population standard deviation is unknown, I use a one-sample t confidence interval.

The standard error is

$$SE = \frac{s}{\sqrt{n}} = \frac{13.09}{\sqrt{25}} \approx 2.62.$$

The critical value for a 90% confidence interval with 24 degrees of freedom is:

```
qt(0.95, df = 24)
```

```
## [1] 1.710882
```

Thus,

$$t^* \approx 1.711.$$

The confidence interval is calculated as

$$\bar{x} \pm t^* SE$$

$$98.44 \pm 1.711(2.62)$$

$$98.44 \pm 4.48.$$

Therefore, the 90% confidence interval is

$$(93.96, 102.92).$$

Thus, based on the sample, the true mean IQ of students in the school is estimated to lie between approximately 93.96 and 102.92 with 90% confidence.

Question 1.2: Hypothesis Test

The hypotheses are

$$H_0 : \mu = 100$$

and

$$H_A : \mu > 100.$$

The test statistic is

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

$$t = \frac{98.44 - 100}{13.09/\sqrt{25}}$$

$$t \approx -0.596.$$

The one-sided hypothesis test is conducted in R as follows:

```
t.test(y, mu = 100, alternative = "greater")

##
## One Sample t-test
##
## data: y
## t = -0.59574, df = 24, p-value = 0.7215
## alternative hypothesis: true mean is greater than 100
## 95 percent confidence interval:
##  93.95993      Inf
## sample estimates:
## mean of x
##      98.44
```

The one-sided p-value is approximately 0.722.

Since $p > 0.05$, I fail to reject the null hypothesis. There is insufficient evidence at the 5% significance level to conclude that the school's average IQ is greater than 100.

Problem 2

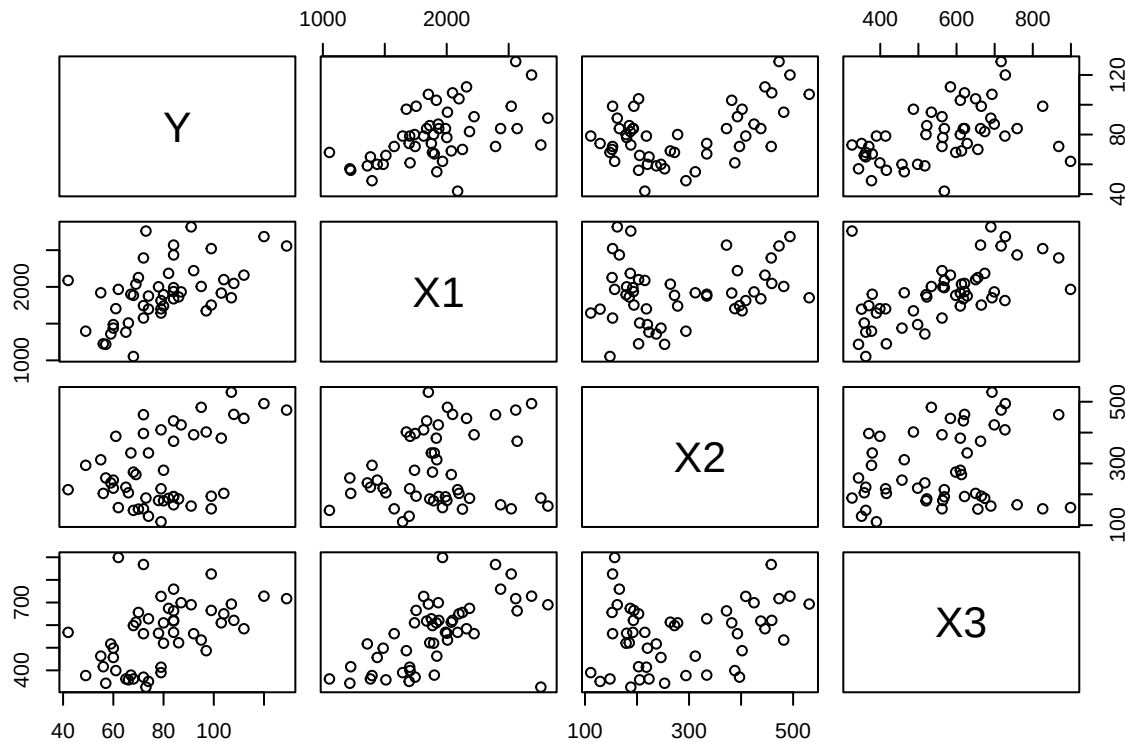
Import the expenditure data into R.

```
expenditure <- read.table(
  "https://raw.githubusercontent.com/ASDS-TCD/StatsI_2026/main/datasets/expenditure.txt",
  header = TRUE
)
```

Question 2(a): Relationships among Y, X1, X2 and X3

Plot the relationships among the four variables.

```
pairs(expenditure[, c("Y", "X1", "X2", "X3")])
```



Calculate the correlations.

```
cor(expenditure[, c("Y", "X1", "X2", "X3")])

##           Y           X1           X2           X3
## Y  1.0000000  0.5317212  0.4482876  0.4636787
## X1 0.5317212  1.0000000  0.2056101  0.5952504
## X2 0.4482876  0.2056101  1.0000000  0.2210149
## X3 0.4636787  0.5952504  0.2210149  1.0000000
```

The scatterplot matrix shows positive relationships between housing-assistance expenditure (Y) and each of the three explanatory variables.

The correlations are approximately:

- Y and $X1$: 0.532
- Y and $X2$: 0.448
- Y and $X3$: 0.464

The strongest relationship with Y is therefore the relationship between housing-assistance expenditure and per-capita personal income ($X1$). All three relationships are positive and moderate in size.

The correlation between $X1$ and $X3$ is also relatively strong at approximately 0.595.

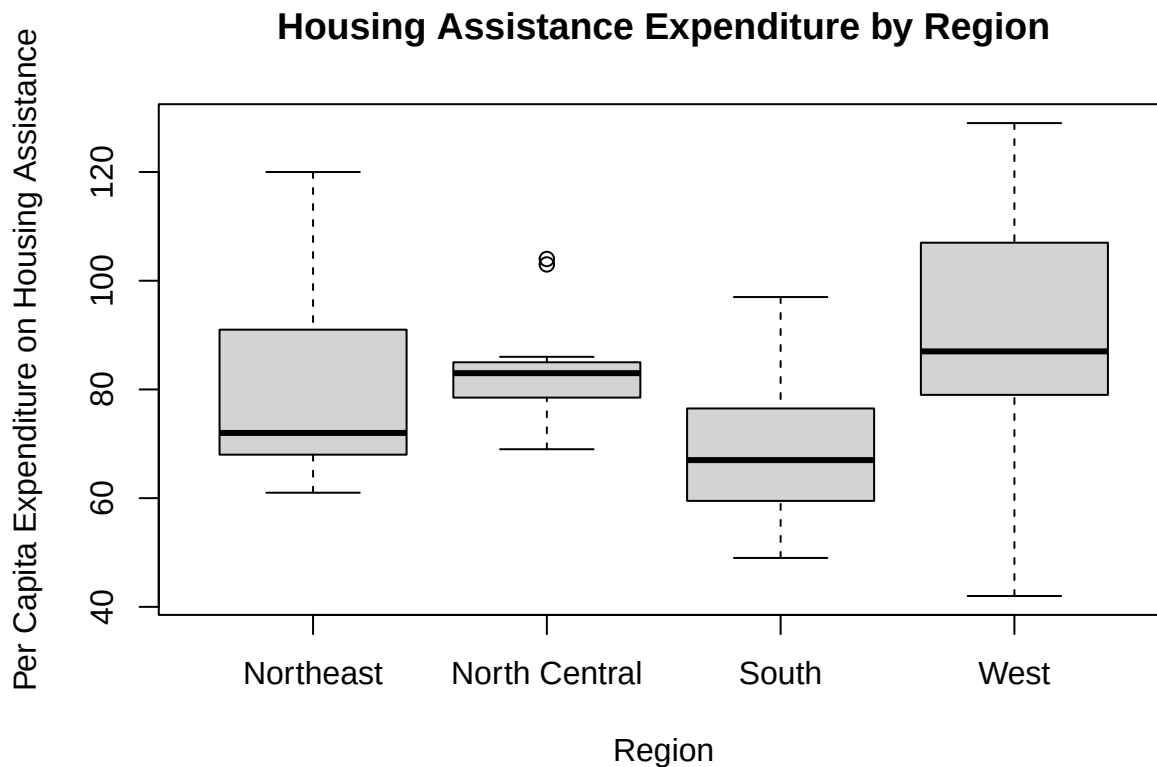
Question 2(b): Housing Assistance Expenditure by Region

First, convert the numerical region codes into labelled categories.

```
expenditure$RegionFactor <- factor(
  expenditure$Region,
  levels = c(1, 2, 3, 4),
  labels = c("Northeast", "North Central", "South", "West")
)
```

Plot housing-assistance expenditure by region.

```
boxplot(
  Y ~ RegionFactor,
  data = expenditure,
  xlab = "Region",
  ylab = "Per Capita Expenditure on Housing Assistance",
  main = "Housing Assistance Expenditure by Region"
)
```



Calculate the mean expenditure for each region.

```
aggregate(Y ~ RegionFactor, data = expenditure, FUN = mean)
```

```
##   RegionFactor      Y
## 1   Northeast 79.44444
## 2 North Central 83.91667
## 3      South 69.18750
## 4       West 88.30769
```

The average per-capita housing-assistance expenditures are approximately:

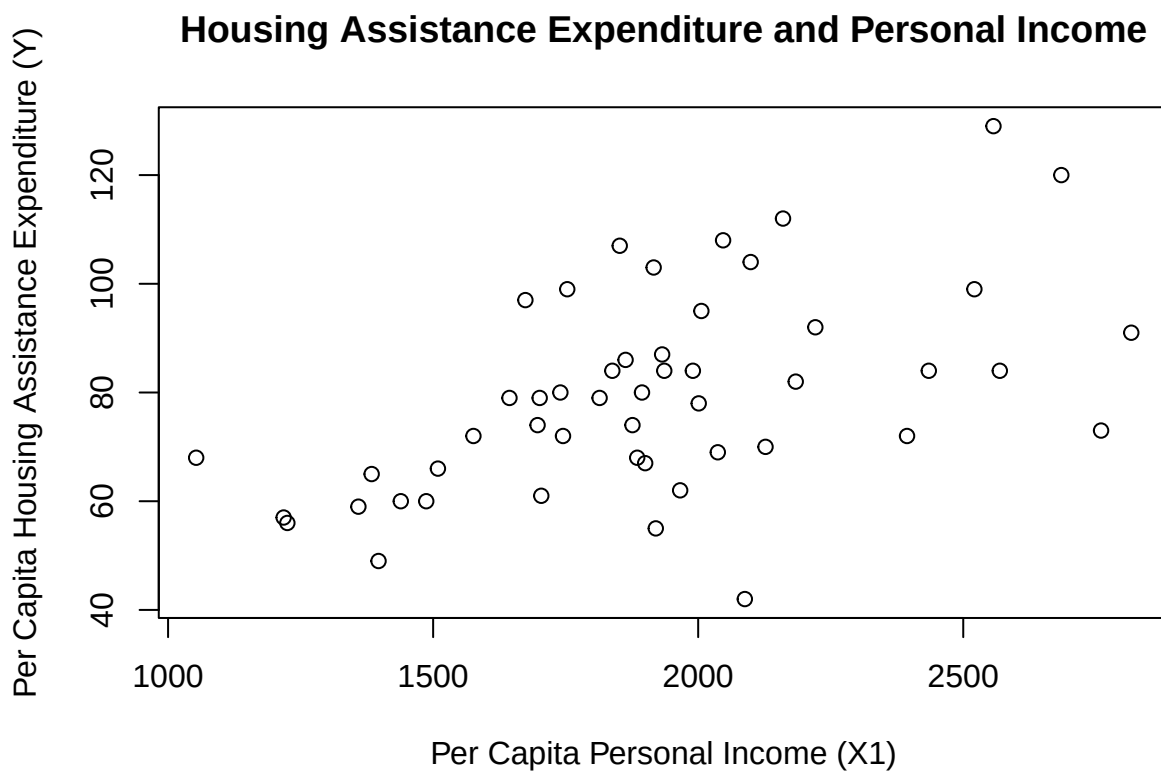
- Northeast: 79.44
- North Central: 83.92
- South: 69.19
- West: 88.31

The West has the highest average per-capita expenditure on housing assistance.

Question 2(c): Relationship between Y and X1

Plot the relationship between per-capita personal income (X_1) and housing-assistance expenditure (Y).

```
plot(
  expenditure$X1,
  expenditure$Y,
  xlab = "Per Capita Personal Income (X1)",
  ylab = "Per Capita Housing Assistance Expenditure (Y)",
  main = "Housing Assistance Expenditure and Personal Income"
)
```



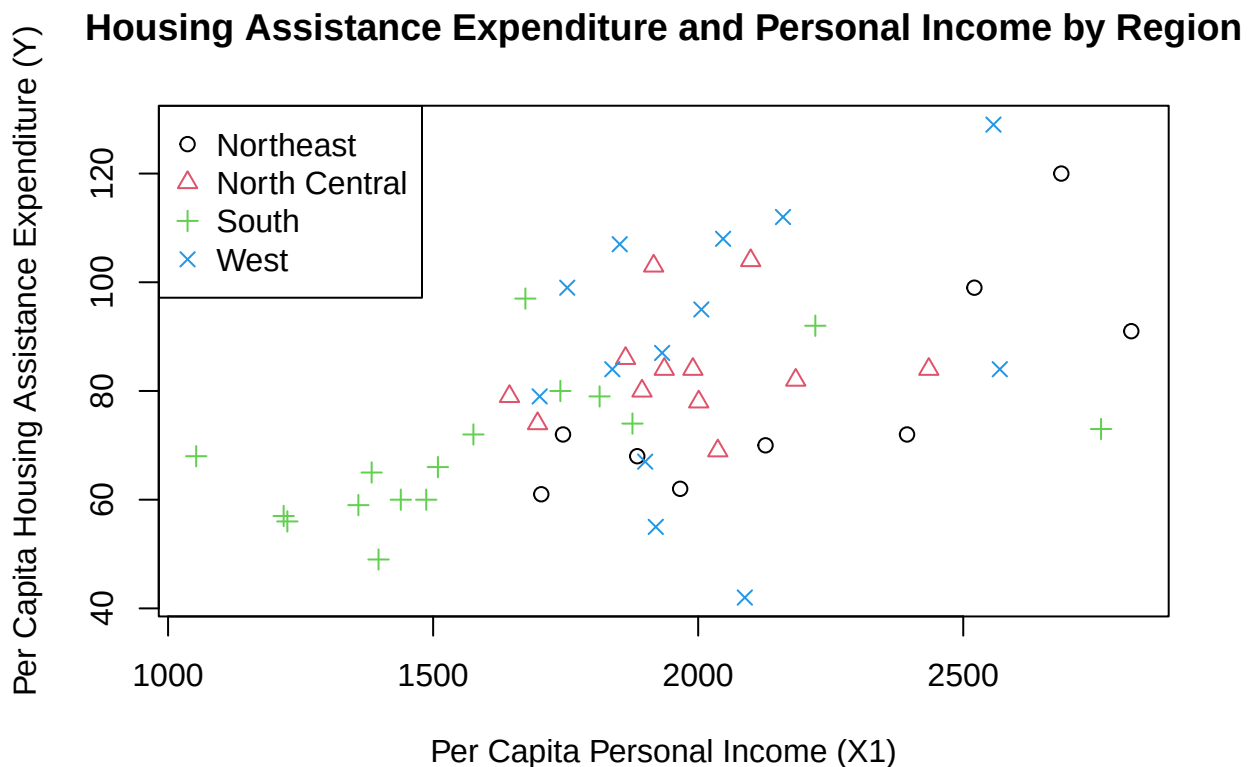
The scatterplot shows a moderate positive relationship between per-capita personal income (X_1) and per-capita housing-assistance expenditure (Y). States with higher personal income tend, on average, to have higher expenditure on housing assistance, although there is considerable variation around this pattern.

Relationship by Region

Reproduce the plot while distinguishing regions using different symbols and colours.

```
plot(
  expenditure$X1,
  expenditure$Y,
  pch = expenditure$Region,
  col = expenditure$Region,
  xlab = "Per Capita Personal Income (X1)",
  ylab = "Per Capita Housing Assistance Expenditure (Y)",
  main = "Housing Assistance Expenditure and Personal Income by Region"
)

legend(
  "topleft",
  legend = c("Northeast", "North Central", "South", "West"),
  pch = 1:4,
  col = 1:4
)
```



When the observations are distinguished by region, Southern states tend to be concentrated toward the lower-income and lower-expenditure part of the graph. Northeastern states are more common at the higher-income end, North Central states tend to cluster around the middle of the distribution, and Western states are more dispersed.